

4. **PC Registers:** PC Registers store the address of the current execution instruction of a thread. Each thread has separate PC Registers.
5. **Native Method Stacks:** A separate native stack is created for each thread, which stores native method information.

Execution Engine

The Execution engine is responsible for executing the bytecode of the compiled Java program. It reads the bytecode instructions and uses data and information present in various memory areas to execute the instructions. It can be divided into three parts:

1. **Interpreter:** This interprets the bytecode line by line and executes it. However, when a method is called multiple times, each time the interpretation process is required, which can lead to inefficiencies.
2. **Just-In-Time (JIT) Compiler:** This is used to improve the efficiency of the interpreter. It compiles the entire bytecode and converts it to native code. When the interpreter sees repeated method calls, the JIT provides the direct native code for that part, which improves efficiency by eliminating the need for re-interpretation.
3. **Garbage Collector:** This component is responsible for destroying unreferenced objects, which helps to manage memory more efficiently. For more information on Garbage Collection, please refer to Garbage Collection.

Java Native Interface (JNI)

JNI (Java Native Interface) is an interface that allows Java programs to interact with native libraries written in languages such as C or C++. It provides the necessary mechanisms for JVM to call C/C++ libraries and vice versa, which may be required for specific hardware or operating systems.

Differences between JDK, JRE and JVM

The Java Development Kit (JDK) is a software development environment used for creating Java applications and applets. It comes with a range of tools such as the Java Runtime Environment (JRE), which includes the Java Virtual Machine (JVM) and core classes necessary for running a Java application. Additionally, the JDK provides a compiler (javac), an archiver (jar), and a documentation generator (Javadoc), among other tools, for Java development. The JRE is the minimum requirement for executing a Java application, and it includes the JVM, core classes, and supporting files. The JVM is responsible for executing Java bytecode and follows a specification that allows for